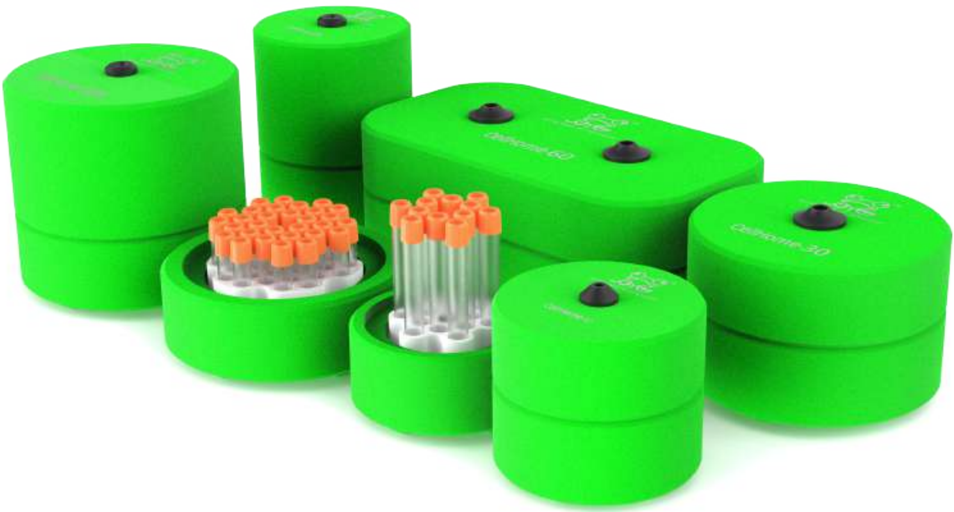
CellHome



User Manual

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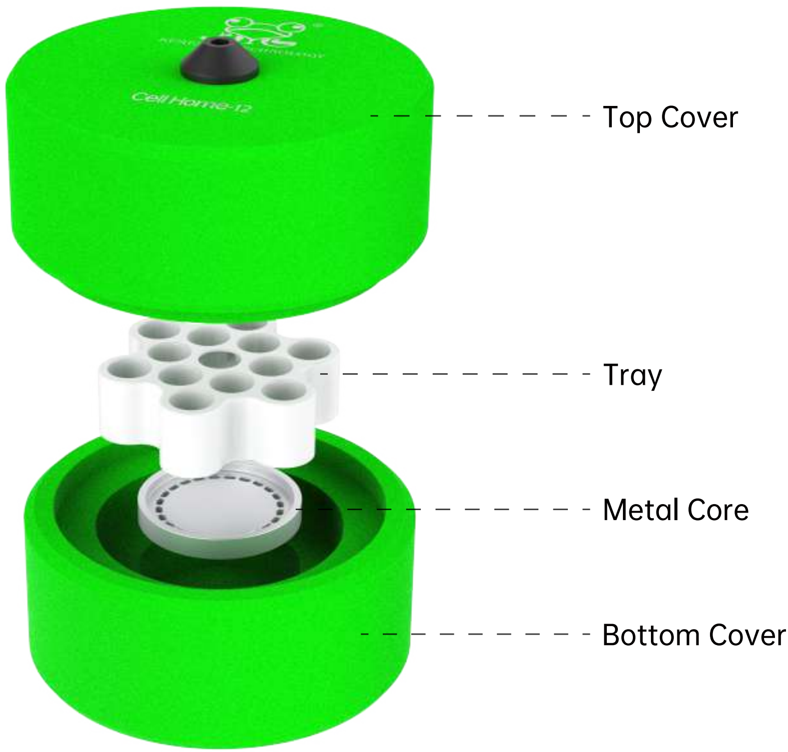
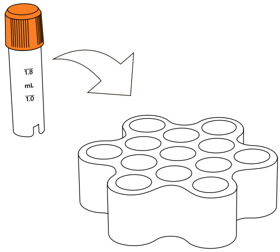
Function description and appearance

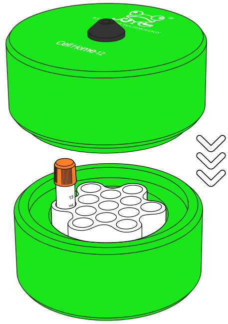
The CellHome series includes five series of cell freezing container:
CellHome-12 (holds 12 cryogenic vials of 1–2 ml),
CellHome-30 (holds 30 cryogenic vials of 1–2 ml),
CellHome-60 (holds 60 cryogenic vials of 1–2 ml),
CellHome-125 (holds 12 cryogenic vials of 5 ml), and
CellHome-305 (holds 30 cryogenic vials of 5 ml).
These units ensure a consistent cooling rate of –1 °C per minute in a low-temperature freezer without the use of alcohol or any other liquid. They are suitable for various cell types, including stem cells, primary cells, and cell lines, making them ideal for most cultured cell applications.
Proven through extensive use with stem cells, primary cells, and cell lines, the CellHome series significantly improves work efficiency and enhances cell viability after preservation.
CellHome achieves an average cooling rate of –1 °C/min by adopting innovative insulation materials and a unique micro-convection design, instead of relying on alcohol or isopropanol.
Easy to use — simply insert the cryogenic vials and place the device in a –80 °C environment.
CellHome provides a safer and more convenient solution for users, ensuring better protection for personnel, environment, and samples while maximizing cell survival.



Instructions for Use

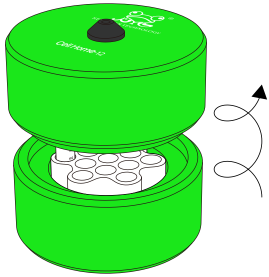


01 Place the frozen samples into the tray.

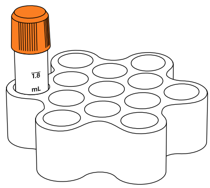


02 Close the lid.

03 Place it in a clean −80 °C freezing environment for at least 3 hours.



04 After the cooling process is complete, rotate to open the (cylindrical) top cover.



05 After the cooling process is complete, remove the samples and transfer them to dry ice or liquid nitrogen.

Tips:

- Before use, dry the tray and cryogenic tubes to prevent them from freezing.
- Ensure that the Core (heat-conducting core) is at room temperature.
- Check whether the cryogenic tubes can be inserted and removed smoothly.
- Secure the lid firmly onto the CellHome unit.
- Place the CellHome vertically into a −80 °C environment, dry ice, or liquid nitrogen, and ensure at least 3 cm of clearance around the unit.
- Freeze for 3~4 hours before transferring the samples to archival storage.
- If CellHome needs to be used continuously, open the top and bottom covers and allow the unit to return to room temperature before reuse.

Experimental Principle

The CellHome series of cell freezing container adopts a high-performance insulating material and a central heat-conducting core (Core) design. By utilizing thermal conduction balance between materials and micro-air convection, it maintains a cooling rate of −1 °C per minute for each cryogenic vial. This linear cooling allows gradual dehydration and crystallization of intracellular water, effectively preventing cell membrane rupture caused by rapid ice crystal formation during freezing, thereby significantly improving cell viability and stability after thawing.

Performance Features

- **Constant cooling rate:** Uniform temperature across all wells with high repeatability.
- **Alcohol-free design:** No need for isopropanol or liquid media, ensuring safe and environmentally friendly operation.
- **High reusability:** Quickly returns to room temperature and can be reused without consumables.
- **Space-saving:** Compact structure compatible with mainstream −80 °C freezers.
- **High compatibility:** Suitable for various types of cryogenic vials (1 mL, 2 mL, 5 mL, etc.).
- **High safety:** Non-toxic and odorless materials that do not contaminate samples or the environment.

Experimental Performance Comparison and Analysis

Comparison Items	Isopropanol-Based series of cell freezing container	CellHome series of cell freezing container
Cooling Medium	Requires addition of isopropanol	No liquid medium required
Operational Safety	Flammable, odorous, and requires waste liquid disposal	Odorless and leak-free
Cooling Rate	Unstable due to degradation with alcohol concentration	Stable linear cooling at an average rate of −1°C/min
Reusability	Requires consumable replenishment	No consumables required; reusable
Maintenance and Cleaning	Difficult to clean due to alcohol residue	Can be washed with water; leaves no residue
Experimental Consistency	Uneven cooling rate across wells	Uniform temperature; reproducible results
Cost Investment	Requires long-term alcohol procurement	One-time purchase; long-term use

Model Classification and Specifications

Picture	Model	Type	Dimensions (mm)	Weight (g)
	CellHome-12	12x1ml or 12x2.0ml Microcentifuge tube or Microtube	Φ118x110	130
	CellHome-30	30x1ml or 30x2.0ml Microcentifuge tube or Microtube	Φ158x116	230
	CellHome-60	60x1ml or 60x2.0ml Microcentifuge tube or Microtube	L303xW177xH125	440
	CellHome-125	12x5.0ml Microcentifuge tube or Microtube	Φ118x163	160
	CellHome-305	30x5.0ml Microcentifuge tube or Microtube	Φ118x165	280

Can control the cell samples'average cooling rate of -1℃/min in -80℃ environment, increase the cell survival rate.

Cleaning and Maintenance

1. Cleaning Method
• Wipe the tray and top cover with warm water and a neutral detergent.
• Rinse thoroughly with purified water and dry completely to keep dry.
• Surface disinfection can be performed using 10% alcohol or bleach solution.
• Do not use high-pressure steam sterilization or soak in strong acids or alkalis.

2. Maintenance Recommendations
• After each use, separate the top cover and tray and let them air dry.
• Avoid direct sunlight or storing near heat sources.
• Use shockproof foam or the original packaging during transportation.
• If the heat-conducting core is loose, cracked, or deformed, stop using the unit and contact after-sales support.

3. Material Characteristics
• The main body is made of high-performance polymer insulating material with excellent frost resistance.
• The core is made of high-thermal-conductivity metal to ensure uniform heat transfer.
• Can withstand temperatures ranging from −96°C to +60°C.
• Free of isopropanol and any harmful substances.

Experimental Recommendations and After-Sales Support

1. Experimental Recommendations
• It is recommended to assess cell recovery rates after cooling to evaluate cryopreservation effectiveness.
• Handle samples in a dry ice environment to prevent temperature rise.
• The exposure time of cryogenic vials should not exceed 30 seconds.
• Adjust freezing duration appropriately for different cell types (generally 3–4 hours).
• Number cryogenic vials corresponding to sample information for easy traceability.
• It is recommended to verify the cooling rate every 3 months to maintain consistency.

2. Storage and Transportation
• Store the product in a dry, cool environment, avoiding high temperature and humidity.
• If not used for an extended period, keep components separated to prevent moisture absorption.
• Avoid severe shaking or dropping during transportation.

3. After-Sales Service
Thank you for choosing the Kemesser CellHome series of cell freezing container.
If you have any questions, please contact our technical staff for assistance:
Phone: 0571-87156759 / 87156259
Email: andy@kemesser.com

Certificate of Conformity：

Certificate of Conformity	
Name:	Programmable Cooling Box
Model：	
Date：	
Test Result：	Qualified
Inspection date：	

Warranty Card:

Warranty:

1.Enjoy ONE year free warranty for the whole machine from the date of purchase;

2.The following situations are not covered by the warranty:

A: Failure and damage caused by not operating correctly according to the manual;

B: Failures and damages caused by self-disassembly and repair;

C: Failures and damages caused by human factors;

D: If the warranty card cannot be presented or product does not have the official sales seal.

Warranty				
No.	Fault cause	Repair dare	Repair man	Remark
1				
2				
3				
4				
5				
6				
7				

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TECHNOLOGY

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